STUDENT'S NAME:._____

LOZI SECONDARY SCHOOL

2019 END OF YEAR FORM THREE EXAMINATIONS

PHYSICS

Monday 22nd July, 2019

PAPER I

(100 marks)

Subject Number: M164/I
Time allowed: 2 hours
8:00 – 10:00 am

Instructions

1. This paper contains **12 printed pages**. Please check.
2. Before beginning, fill in your **NAME** at the top of each page.
3. This paper contains **two** sections, **A** and **B**. In **Section A** there are **ten** short answer questions while in **Section B** there are **three** restricted essay questions.
4. Answer **all** the **thirteen** questions in the spaces provided
5. Use of electronic calculators is allowed.
6. The maximum number of marks for each answer is indicated against each question.
7. In the table provided on this page. **Tick** against the question number you have answered.

Question Number	Tick if answered	Do not write in these columns		Examiner's Name
1				SCK
2				SCK
3				SCK
4				SCK
5				SCK
6				SCK
7				SCK
8				SCK
9				SCK
10				SCK
11				SCK
12				SCK
13				SCK
				SCK

SECTION A [70 Marks]

1. a. State any **two** effects of force on an object.

_____

_____

[2 marks]

b. Why does a person experience difficulty walking in the mud or sand?

_____

_____

_____

[2 marks]

c. Define 'diffusion'.

_____

_____

[1 mark]

d. State any **two** properties of matter that are common between gases and liquids.

_____

_____

[2 marks]

e. **Figure 1** show an experiment set up to demonstrate gas pressure, use it to answer questions.

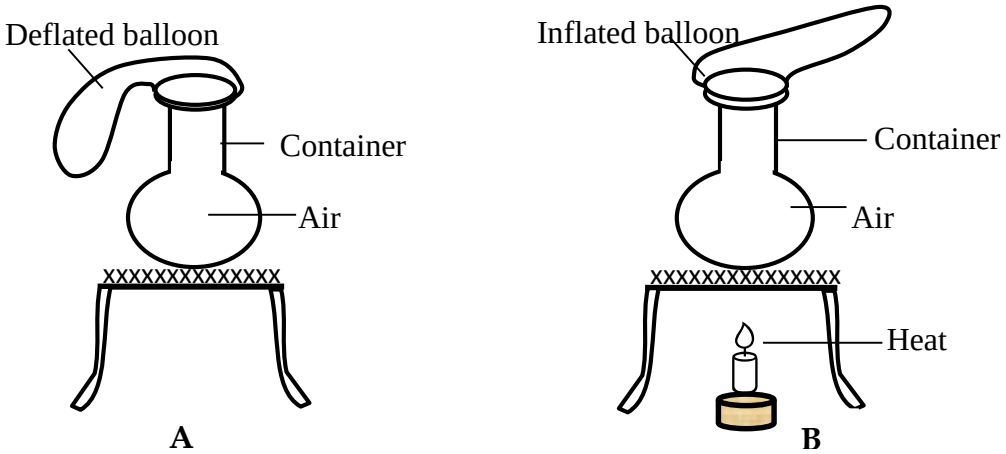

Figure 1

f. Using molecular motion of matter, explain why a balloon in **B** gets inflated after a container is heated.

_____

_____

_____

[3 marks]

g. Apart from increase in temperature mention **any** other factor that can increase pressure of air in container.

_____

_____

[1 mark]

2. a. Mention **two** appliances that use the principle of electrostatics

_____

_____

[2 marks]

b. Explain why mountains remain cold during summers.

_____

_____

_____

[3 marks]

c. Explain how a desert gets too hot during daytime and too cold at night.

_____

_____

_____

[3 Marks]

d. You are provided with a bucket full of sand and a bucket full of water.

i. Which bucket becomes hot faster than the other?

_____

[1 mark]

ii. Give a reason for your answer in **2d** (i).

_____

_____

_____

[2 marks]

3. a. Give any two effects of heat on matter.

_____

_____

[2 marks]

b. Explain why temperature remains constant during the change of state of ice to liquid water.

_____

_____

_____

[2 marks]

4. a. Define the term light.

_____

_____

[1 mark]

b. Mention any **two** characteristics of shadow formed by a point source of light.

_____

_____

[2 marks]

c. Mention any two applications of reflection of light.

_____

_____

[2 marks]

5. a. Define the term shadow.

_____

_____

[1 mark]

b. Explain how shadow of an object is formed.

_____

_____

_____

[2 marks]

6. a. In the **Figure 2** below, the flask is being held just under the surface of water. When the clip is opened, nothing goes into the flask and nothing comes out of it.

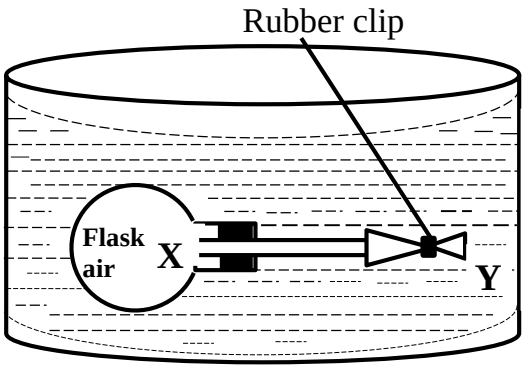

Figure 2

i. How does the pressure at **X** compare with that at **Y**?

_____

_____

[3 marks]

ii. If the flask is pushed further down into the water, water goes into it, but no air leaves it. Why is this?

_____

_____

_____

[3 marks]

c. Two flasks shown in the **Figure 3** below contain equal volumes of carbon dioxide and hydrogen gases. They are both immersed into ice water simultaneously.

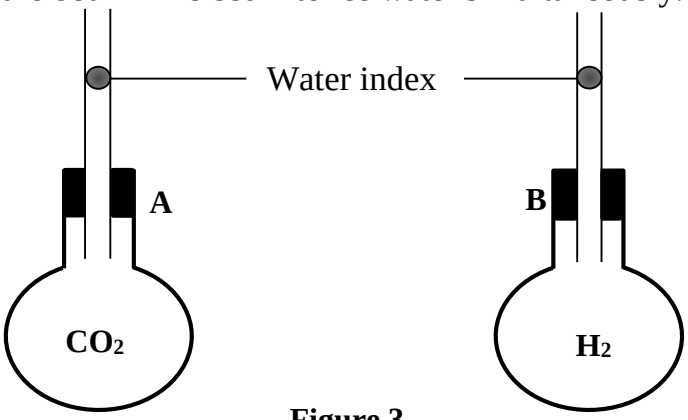

Figure 3

i. What will happen to the water indices in flask **A** and **B**?

_____

_____

[2 marks]

ii. How will the level of indices of **A** and **B** compare?

_____

_____

_____

[3 marks]

7. a. Define "centre of gravity"

_____

_____

[1 mark]

b. State any two ways of increasing mechanical advantage of an inclined plane.

_____

_____

_____

_____

[2 marks]

c. **Figure 4** below is a diagram showing an unbalanced lever, the lengths being in cm.

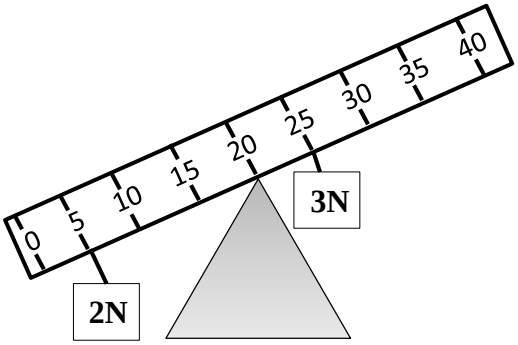

Figure 4

i. On what position on the lever would you place a **1N** weight in order to make it balance?

_____

[1 mark]

8. a. Define scalar quantities.

_____

_____

[1 mark]

b. Give two properties of vector quantities.

_____

_____

[2 marks]

c. Mention one method of adding vectors acting at an angle to each other.

_____

[1 mark]

d. Mention any two examples of a scalar quantity.

_____

_____

[2 marks]

e. Describe how vectors are represented on paper.

_____

_____

[2 marks]

f. Define "resultant vector".

_____

_____

[2 marks]

g. Why is "speed" a scalar quantity while "velocity" a vector quantity?

_____

_____

_____

[2 marks]

9. a. **Figure 5** is diagram showing a piece of card and a coin of equal mass released at the same time from a height of 10m. Use it to answer the questions that follow.

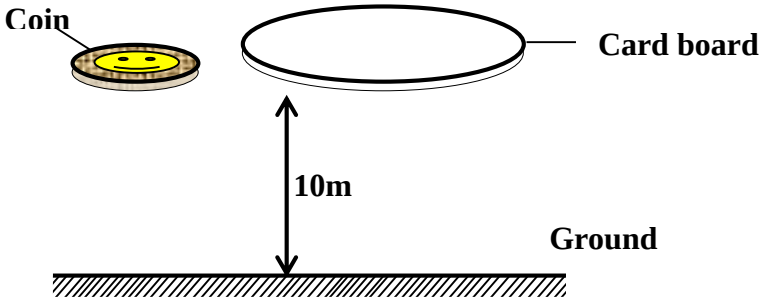

Figure 5

i. Which of the two would reach the ground first? Give a reason for your answer.

_____

_____

[3 marks]

ii. State the three forces which act on each object as it falls.

_____

_____

[3 marks]

iii. Which forces remain constant as the object falls?

_____

[1 mark]

10. a. Define oscillations as applied to:

i. Swing pendulum:

_____

_____

[1 mark]

ii. Cantilever:

_____

_____

[1 mark]

SECTION B [30 marks]

11. a. Explain how a terminal speed is reached by an object falling through a liquid.

_____

_____

_____

_____

[2 marks]

b. Describe how a bimetallic strip regulates the temperature in an electric iron.

_____

_____

_____

_____

[3 marks]

c. Explain how the action of flotation and sinking is achieved in submarine ships.

_____

_____

_____

[3 marks]

12. a. Explain how in a glass thermometer works to determine temperature of a substance.

_____

_____

_____

_____

_____

[3 marks]

b. Describe an experiment that could be carried out in order to show that liquid pressure increases with depth.

_____

_____

_____

_____

_____

_____

_____

_____

_____

[4 marks]

c. Explain why a rotten egg floats on water while fresh egg sinks.

_____

_____

_____

_____

_____

[2 marks]

13. a. **Figure 6** is a diagram showing a set up of an experiment investigating the fluid resistance to motion of objects. The ball bearings were released into the liquids simultaneously. Use it to answer the questions that follow.

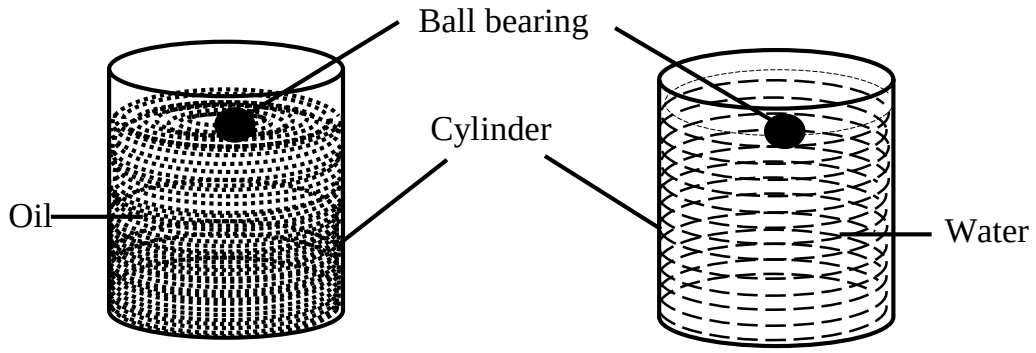

Figure 6

(i) Which ball bearing reached the bottom of the cylinder first?

_____

[1 mark]

(ii) Mention any **two** variables that were kept constant in the experiment.

_____

_____

[1 mark]

b. **Table 1** shows results of an experiment. Use it to answer the questions that follow.

Table 1

Length of wire (cm)	**Voltage (V)**	**Current (A)**	**Voltage / current**
20	0.6	0.20	3
40	0.6	0.10	6
60	0.6	0.06	10
80	0.6	0.05	

(i) Work out the missing value in the last column of the table. **[1 mark]**

(ii) What factor was being investigated in this experiment?

_____

[1 mark]

(iii) What conclusion can you draw from this experiment?

_____

_____

[2 marks]

c. A student set up an experiment to demonstrate the dispersion of white light.

(i) Draw a labelled diagram of the setup of the apparatus the student may have used.

[2 marks]

(ii) What name is given to the band of colors produced when white light is dispersed?

_____

[1 mark]

d. **Figure 7** is a diagram showing a device which switches on and off electricity automatically in an electric appliance. Use it to answer the questions that follow.

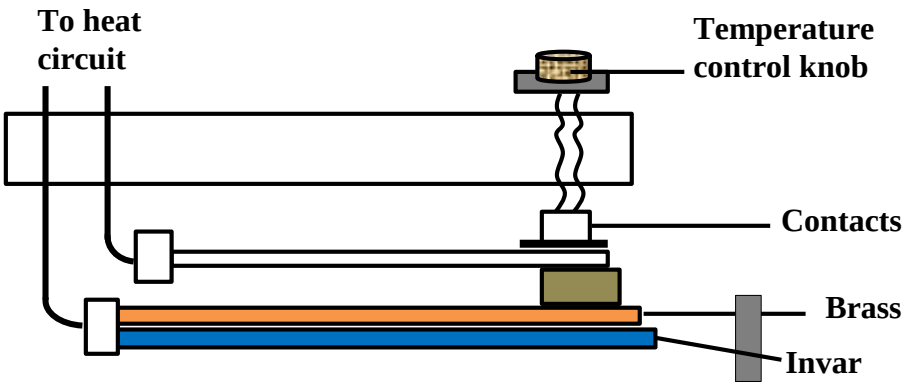

Figure 7

(i) Name the device.

_____

[1 mark]

(ii) Explain how the device works.

_____

_____

_____

_____

[2 marks]

(iii) Mention any two electrical appliances which use the device.

_____

_____

[1 mark]

END OF QUESTION PAPER

NB: This paper contains 12 pages.